MASON STASZEL

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EDUCATION

University of Michigan | *College of Engineering* — Ann Arbor, MI | **Expected May 2027**

Bachelor of Science in Engineering (B.S.E.) in Computer Engineering — **Current GPA: 3.62**

Relevant Coursework: Computer Architecture, Embedded Systems, Computer Networks, Data Structures & Algorithms, Computer Organization, Cybersecurity, Logic Design, Electronic Circuits, Signals & Systems.

Planned (Winter 2027): Operating Systems, Advanced Embedded Systems

TECHNICAL SKILLS

- **Languages:** C/C++, Verilog/SystemVerilog, Python, Java, MATLAB, Bash/Shell, LaTeX
- **Hardware & Architectures:** ARM Cortex-M4 (Infineon XMC4700), RISC Architecture, FPGA, BIOS/UEFI Firmware, NUMA Tuning
- **Systems & Infrastructure:** Linux/Unix, Docker, Portainer, TrueNAS Scale, Tailscale, Windows Server
- **Developer Tools & Test:** Git/GitHub, llama.cpp, Claude Code, Wireshark, Logic Analyzers, Oscilloscopes, LTSpice

ENGINEERING EXPERIENCE

Rocket | *Systems Engineering Intern* — Detroit, MI | **May 2026 - August 2026**

- Benchmarked and tuned on-premise CPU LLM inference using llama.cpp across multi-socket enterprise servers, configuring thread pinning, NUMA memory affinity, and GGUF quantization to optimize token throughput and reduce decode latency.
- Migrated existing Windows setup scripts to configure a standardized Ubuntu development VM for macOS, maintaining corporate endpoint monitoring compliance across local developer machines.
- Collaborated within an Agile engineering workflow, utilizing Azure DevOps (ADO) to track operational tasks and participating in standups to coordinate routine Windows Server update schedules and maintenance.

TECHNICAL PROJECTS

Home Server & Self-Hosted Infrastructure | *Personal Project* — TrueNAS Scale, Docker, Tailscale | **2020 - Present**

- Maintained a 24/7 personal homelab server running TrueNAS Scale with 12 TB storage in a RAIDZ1 ZFS configuration, managing automated snapshot replication and data scrubbing.
- Deployed 8+ containerized microservices via Docker and Portainer, establishing encrypted zero-trust remote access through a Tailscale mesh network without opening external firewall ports.
- Set up local DNS routing and reverse proxy rules to manage internal hostnames, port mapping, and automatic SSL/TLS certificate renewal for hosted web interfaces.

Decoupled Superscalar Processor | *University of Michigan* — SystemVerilog, Computer Architecture | **Fall 2026 (In Progress)**

- Collaborated in a 5-person team to architect and synthesize a 2-way superscalar processor, driving weekly development milestones through structured interface planning, architectural reviews, and technical progress reporting.
- Co-developed the decoupled pipeline and micro-op queue through live team programming and continuous simulation checkpoints, coordinating module integration across frontend and backend stages to debug hazards and verify synthesized hardware correctness.
- Triaged simulation failures alongside teammates by inspecting waveform traces and comparing cycle-by-cycle writeback logs against reference architectural models to isolate forwarding hazards and branch squash bugs.

Adaptive Video Streaming CDN & HTTP Proxy | *University of Michigan* — C++, POSIX Sockets | **Fall 2026**

- Developed a multi-client HTTP proxy in C++ using select() I/O multiplexing to intercept MPEG-DASH video streams, parse XML manifest files (pugixml), and rewrite segment request URIs on-the-fly.
- Estimated per-client throughput using an exponentially weighted moving average to dynamically select supported bitrates, and built a CDN load balancer implementing round-robin and shortest-path (Dijkstra) geographic server routing.

Web Application Vulnerability & Penetration Testing | *University of Michigan* — Python, SQL | **Winter 2026**

- Executed proof-of-concept exploits (SQLi, XSS, CSRF) within an isolated environment to demonstrate data exfiltration vectors, broken authentication, and unauthorized backend execution.
- Authored Python scripts to automate payload delivery, evaluating session token leakage and validating backend input sanitization.

LEADERSHIP & ACTIVITIES

Boy Scouts of America | *Eagle Scout* — Brighton, MI | **October 2023**

- Directed a community improvement renovation project; coordinated 10+ volunteers across 100+ cumulative service hours to deliver the project on time and within budget.

Brighton High School Aquatics | *Lifeguard* — Brighton, MI | **May 2022 - June 2024**

- Supervised community aquatic facility operations, enforced safety protocols for 100+ daily visitors, and resolved patron incidents calmly and professionally.